

101. Let $y = y(x)$ be the solution of the differential equation $(xy - 5x^2\sqrt{1+x^2})dx + (1+x^2)dy = 0$, $y(0) = 0$. Then $y(\sqrt{3})$ is equal to :

माना कि $y = y(x)$ अवकल समीकरण $(xy - 5x^2\sqrt{1+x^2})dx + (1+x^2)dy = 0$, $y(0) = 0$ का हल है। तब $y(\sqrt{3})$ बराबर है :

- (A) $2\sqrt{2}$ (B) $\sqrt{\frac{14}{3}}$ (C) $\frac{5\sqrt{3}}{2}$ (D) $\sqrt{\frac{15}{2}}$

Ans. C

Sol. I.F. = $\int \frac{x dx}{1+x^2} = e^{\frac{1}{2}\ln(1+x^2)} = \sqrt{1+x^2}$

Solution of differential equation

$$y \cdot \sqrt{1+x^2} = \int \sqrt{1+x^2} \cdot \frac{5x^2}{\sqrt{1+x^2}} dx + C$$

$$y\sqrt{1+x^2} = \frac{5x^3}{3} + C$$

when $x = 0$, $y = 0$ then $C = 0$

$$\text{So, } y = \frac{5x^3}{3\sqrt{1+x^2}}$$

$$\therefore y(\sqrt{3}) = \frac{5 \cdot 3\sqrt{3}}{3 \cdot 2} = \frac{5\sqrt{3}}{2}$$

102. Let for some function $y = f(x)$, $\int_0^x tf(t) dt = x^2 f(x)$, $x > 0$ and $f(2) = 3$. Then $f(6)$ is equal to :

माना कि किसी फलन $y = f(x)$ के लिए, $\int_0^x tf(t) dt = x^2 f(x)$, $x > 0$ और $f(2) = 3$ है। तब $f(6)$ बराबर है :

- (A) 1 (B) 3 (C) 6 (D) 2

Ans. A

Sol. $\int_0^x tf(t) dt = x^2 f(x)$, $x > 0$

Differentiating both side w.r.t. x

$$xf(x) = x^2 f'(x) + 2xf(x)$$

$$-xf(x) = x^2 f'(x)$$

$$\int \frac{f'(x)}{f(x)} dx = \int \frac{-1}{2} dx$$

$$\log f(x) = -\log x + \log c$$

$$f(x) = \frac{c}{x}$$

$$f(2) = 3 \Rightarrow 3 = \frac{c}{2} \Rightarrow c = 6$$

$$f(x) = \frac{6}{x}$$

$$f(6) = 1 \quad \therefore (1)$$

103. Let $y = y(x)$ be the solution of the differential equation

$\cos x (\log_e (\cos x))^2 dy + (\sin x - 3y \sin x \log_e (\cos x)) dx = 0, x \in \left(0, \frac{\pi}{2}\right)$. If $y\left(\frac{\pi}{4}\right) = \frac{-1}{\log_e 2}$, then $y\left(\frac{\pi}{6}\right)$ is equal to :

माना कि $y = y(x)$, अवकल समीकरण $\cos x (\log_e (\cos x))^2 dy + (\sin x - 3y \sin x \log_e (\cos x)) dx = 0, x \in \left(0, \frac{\pi}{2}\right)$ का

हल है। यदि $y\left(\frac{\pi}{4}\right) = \frac{-1}{\log_e 2}$ है, तो $y\left(\frac{\pi}{6}\right)$ बराबर है :

(A) $-\frac{1}{\log_e (4)}$ (B) $\frac{2}{\log_e (3) - \log_e (4)}$ (C) $\frac{1}{\log_e (4) - \log_e (3)}$ (D) $\frac{1}{\log_e (3) - \log_e (4)}$

Ans. D

Sol. $\cos x (\ln \cos x)^2 dy = (3y \sin x \ln \cos x - \sin x) dx$

$$\frac{dy}{dx} = \frac{3y \sin x \ln \cos x}{\cos x (\ln \cos x)^2} - \frac{\tan x}{(\ln \cos x)^2}$$

$$\frac{dy}{dx} - \frac{3 \tan x}{\ln \cos x} = -\frac{\tan x}{(\ln \cos x)^2}$$

$$\text{I.F.} = e^{\int \frac{3 \tan x}{\ln \cos x} dx}$$

$$\ln \cos x = t$$

$$\frac{1}{\cos x} \times -\sin x dx = dt$$

$$-\tan x dx = dt$$

$$\text{I.F.} = e^{\int \frac{3dt}{t}} = e^{3 \ln t} = t^3 = (\ln \cos x)^3$$

$$y \times (\ln \cos x)^3 = -\int \tan x (\ln \cos x) dx$$

$$\ln \cos x = t$$

$$-\tan x dx = dt$$

$$y (\ln \cos x)^3 = \frac{t^2}{2} + C$$

$$y (\ln \cos x)^3 = \frac{(\ln \cos x)^2}{2} + C$$

$$C = 0$$

$$y = \frac{1}{2 \ln \cos x}$$

$$y = \frac{1}{2 \ln \frac{\sqrt{3}}{2}}$$

$$y = \frac{1}{2 (\ln \sqrt{3} - \ln 2)}$$

$$y = \frac{1}{\ln 3 - \ln 4}$$

104. If for solution curve $y = f(x)$ of the differential equation $\frac{dy}{dx} + (\tan x)y = \frac{2 + \sec x}{(1 + 2 \sec x)^2}$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{10}$, then $f\left(\frac{\pi}{4}\right)$ is equal to :

यदि अकवल समीकरण $\frac{dy}{dx} + (\tan x)y = \frac{2 + \sec x}{(1 + 2 \sec x)^2}$, $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ के हल वक्र $y = f(x)$ के लिए $f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{10}$ है, तो

$f\left(\frac{\pi}{4}\right)$ बराबर है :

(A) $\frac{4-\sqrt{2}}{14}$

(B) $\frac{5-\sqrt{3}}{2\sqrt{2}}$

(C) $\frac{9\sqrt{3}+3}{10(4+\sqrt{3})}$

(D) $\frac{\sqrt{3}+1}{10(4+\sqrt{3})}$

Ans. B

Sol. If $e^{\int \tan x dx} = e^{\ln(\sec x)} = \sec x$

$$\therefore y \cdot \sec x = \int \left\{ \frac{2 + \sec x}{(1 + 2 \sec x)^2} \right\} \sec x dx$$

$$= \int \frac{2 \cos x + 1}{(\cos x + 2)^2} dx$$

Let $\cos x = \frac{1-t^2}{1+t^2}$

$$= \int \frac{2 \left(\frac{1-t^2}{1+t^2} \right) + 1}{\left(\frac{1-t^2}{1+t^2} + 2 \right)^2} 2dt$$

$$= \int \frac{2 - 2t^2 + 1 + t^2}{(1 - t^2 + 2 + 2t^2)^2} \times 2dt$$

$$= 2 \int \frac{3 - t^2}{(t^2 + 3)^2} dt$$

Let $t + \frac{3}{t} = u$

$$\left(1 - \frac{3}{t^2} \right) dt = du$$

$$= -2 \int \frac{du}{u^2}$$

$$y \cdot (\sec x) = \frac{2}{u} + c$$

$$y \cdot \sec x = \frac{2}{t + \frac{3}{t}} + c \quad \text{_____ (i)}$$

$$\text{At } x = \frac{\pi}{3}, t = \tan \frac{x}{2} = \frac{1}{\sqrt{3}}$$

$$2 \cdot \frac{\sqrt{3}}{10} = \frac{2}{\frac{1}{\sqrt{3}} + 3\sqrt{3}} + c$$

$$2 \cdot \frac{\sqrt{3}}{10} = \frac{2\sqrt{3}}{10} + c \Rightarrow C = 0$$

$$\text{At } x = \frac{\pi}{4}, t = \tan \frac{x}{2} = \sqrt{2} - 1$$

$$\therefore y \cdot \sqrt{2} = \frac{2}{\sqrt{2} - 1 + \frac{3}{\sqrt{2} - 1}}$$

$$y \cdot \sqrt{2} = \frac{2(\sqrt{2} - 1)}{6 - 2\sqrt{2}}$$

$$y = \frac{\sqrt{2}(\sqrt{2} - 1)}{2(3 - \sqrt{2})} = \frac{1}{\sqrt{2}} \times \frac{2\sqrt{2} - 1}{7}$$

$$= \frac{4 - \sqrt{2}}{14}$$

105. If the solution $y = y(x)$ of the differential equation $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ satisfies $y(-1) = -\frac{\pi}{4}$, then $y(0)$ is equal to :

यदि अवकल समीकरण $(x^4 + 2x^3 + 3x^2 + 2x + 2)dy - (2x^2 + 2x + 3)dx = 0$ का हल $y = y(x)$, $y(-1) = -\frac{\pi}{4}$ को संतुष्ट करता है, तो $y(0)$ बराबर है :

- (A) 0 (B) $-\frac{\pi}{12}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

Ans. D

Sol. $\int dy = \int \frac{(2x^2 + 2x + 3)}{x^4 + 2x^3 + 2x + 2} dx$

$$y = \int \frac{(2x^2 + 2x + 3)}{(x^2 + 1)(x^2 + 2x + 2)} dx$$

$$y = \int \frac{dx}{x^2 + 2x + 2} + \int \frac{dx}{x^2 + 1}$$

$$y = \tan^{-1}(x + 1) + \tan^{-1}x + C$$

$$y(-1) = -\frac{\pi}{4}$$

$$-\frac{\pi}{4} = 0 - \frac{\pi}{4} + C \Rightarrow C = 0$$

$$\Rightarrow y = \tan^{-1}(x + 1) + \tan^{-1}x$$

$$y(0) = \tan^{-1} 1 = \frac{\pi}{4}$$

106. Let $y = y(x)$ be the solution of the differential equation $(x^2 + 4)^2 dy + (2x^3 y + 8xy - 2) dx = 0$. If $y(0) = 0$, then $y(2)$ is equal to :

माना $y = y(x)$ अवकल समीकरण $(x^2 + 4)^2 dy + (2x^3 y + 8xy - 2) dx = 0$ का हल है। यदि $y(0) = 0$ है, तो $y(2)$ बराबर है:

(A) $\frac{\pi}{32}$

(B) $\frac{\pi}{16}$

(C) 2π

(D) $\frac{\pi}{8}$

Ans. A

Sol. $\frac{dy}{dx} + \left(\frac{2x}{x^2 + 4} \right) y = \frac{2}{(x^2 + 4)^2}$

$$\text{I.F.} = e^{\int \frac{2x}{x^2 + 4} dx} = e^{\ln(x^2 + 4)} = x^2 + 4$$

$$\text{solution of differential equation } y(x^2 + 4) = \int \frac{2}{x^2 + 4} dx \Rightarrow y(x^2 + 4) = \tan^{-1} \left(\frac{x}{2} \right) + C$$

$$\text{But } y(0) = 0 \therefore c = 0$$

$$y(x^2 + 4) = \tan^{-1} \left(\frac{x}{2} \right)$$

$$\text{at } x = 2 \rightarrow y = \frac{\pi/4}{8} = \frac{\pi}{32}$$

107. The differential equation of the family of circles passing through the origin and having center at the line $y = x$ is:

मूल बिन्दु से होकर जाने वाले तथा रेखा $y = x$ पर केन्द्र के वृत्त कुल का अवकल समीकरण है :

(A) $(x^2 - y^2 + 2xy)dx = (x^2 - y^2 - 2xy)dy$

(B) $(x^2 + y^2 + 2xy)dx = (x^2 - y^2 + 2xy)dy$

(C) $(x^2 + y^2 - 2xy)dx = (x^2 + y^2 + 2xy)dy$

(D) $(x^2 + y^2 + 2xy)dx = (x^2 + y^2 - 2xy)dy$

Ans. A

Sol. $C \equiv x^2 + y^2 + gx + gy = 0 \dots (1)$

$$2x + 2yy' + g + gy' = 0$$

$$g = -\left(\frac{2x + 2yy'}{1 + y'}\right)$$

put in (1)

$$x^2 + y^2 - \left(\frac{2x + 2yy'}{1 + y'}\right)(x + y) = 0$$

$$(x^2 - y^2 - 2xy)y' = x^2 - y^2 + 2xy$$

108. Let $y = y(x)$ be the solution of the differential equation $(2x \log_e x) \frac{dy}{dx} + 2y = \frac{3}{x} \log_e x, x > 0$ and $y(e^{-1}) = 0$.

Then, $y(e)$ is equal to :

माना $y = y(x)$ अवकल समीकरण $(2x \log_e x) \frac{dy}{dx} + 2y = \frac{3}{x} \log_e x, x > 0$ तथा $y(e^{-1}) = 0$ का हल है, तो $y(e)$ बराबर है:

(A) $-\frac{3}{2e}$

(B) $-\frac{3}{e}$

(C) $-\frac{2}{e}$

(D) $-\frac{2}{3e}$

Ans. B

Sol. $(1 + x^2) \cdot \frac{dy}{dx} + y = e^{\tan^{-1} x}$

Let $\tan^{-1} x = t$

$$\frac{1}{1 + x^2} \cdot dx = dt$$

$$\frac{dy}{dt} + y = e^t$$

$$\text{If } = e^{\int 1 \cdot dt} = e^t$$

$$y \cdot e^t = \frac{e^{2t}}{2} + C$$

$$y \cdot e^{\tan^{-1} x} = \frac{e^{2 \tan^{-1} x}}{2} + C$$

$$x = 1 \quad y = 0 \quad C = -\frac{e^{\pi/2}}{2}$$

$$y \cdot \tan^{-1} x = \frac{e^{2 \tan^{-1} x}}{2} - \frac{e^{\pi/2}}{2}$$

$$x = 0 \Rightarrow y = \frac{1}{2}(1 - e^{\pi/2})$$

109. Let $y = y(x)$ be the solution of the differential equation $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$, $y(1) = 0$. Then $y(0)$ is :

माना $y = y(x)$ अवकल समीकरण $(1 + x^2) \frac{dy}{dx} + y = e^{\tan^{-1} x}$, $y(1) = 0$ का हल है, तो $y(0)$ है :

(A) $\frac{1}{2}(1 - e^{\pi/2})$ (B) $\frac{1}{4}(e^{\pi/2} - 1)$ (C) $\frac{1}{4}(1 - e^{\pi/2})$ (D) $\frac{1}{2}(e^{\pi/2} - 1)$

Ans. A

Sol. $(1 + x^2) \cdot \frac{dy}{dx} + y = e^{\tan^{-1} x}$

Let $\tan^{-1} x = t$

$$\frac{1}{1 + x^2} \cdot dx = dt$$

$$\frac{dy}{dt} + y = e^t$$

$$\text{If } = e^{\int 1 \cdot dt} = e^t$$

$$y \cdot e^t = \frac{e^{2t}}{2} + C$$

$$y \cdot e^{\tan^{-1} x} = \frac{e^{2 \tan^{-1} x}}{2} + C$$

$$x = 1 \quad y = 0 \quad C = -\frac{e^{\pi/2}}{2}$$

$$y \cdot \tan^{-1} x = \frac{e^{2 \tan^{-1} x}}{2} - \frac{e^{\pi/2}}{2}$$

$$x = 0 \Rightarrow y = \frac{1}{2}(1 - e^{\pi/2})$$

110. Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ represents a circle passing through origin. Then the radius of this circle is :

मान लीजिए कि अवकल समीकरण $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ का हल मूल बिन्दु से गुजरने वाले वृत्त को दर्शाता है, तो इस वृत्त की त्रिज्या है :

- (A) $\frac{\sqrt{17}}{2}$ (B) 2 (C) $\sqrt{17}$ (D) $\frac{1}{2}$

Ans. A

Sol. $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - y(2\alpha + \beta) + 4\alpha}$

$$\beta x dy - (2\alpha + \beta) y dy + 4\alpha dy = (2 + \alpha) x dx - \beta y dx + 2 dx$$

$$\beta (x dy + y dx) - (2\alpha + \beta) y dy + 4\alpha dy = (2 + \alpha) x dx + 2 dx$$

$$\beta xy - \frac{(2\alpha + \beta)y^2}{2} + 4\alpha y = \frac{(2 + \alpha)x^2}{2}$$

$$\Rightarrow \beta = 0 \text{ for this to be circle}$$

$$(2 + \alpha) \frac{x^2}{2} + \alpha y^2 + 2x - 4\alpha y = 0$$

$$\text{coeff. of } \begin{matrix} x^2 \\ y^2 \end{matrix} > 2 + a = 2a$$

$$\Rightarrow \alpha = 2$$

$$\text{i.e. } 2x^2 + 2y^2 + 2x - 8y = 0$$

$$x^2 + y^2 + x - 4y = 0$$

$$rd = \sqrt{\frac{1}{4} + 4} = \frac{\sqrt{17}}{2}$$

111. Let $y = y(x)$ be the solution of the differential equation $(1 + y^2)e^{\tan x} dx + \cos^2 x (1 + e^{2\tan x}) dy = 0$, $y(0) = 1$. Then

$$y\left(\frac{\pi}{4}\right) \text{ is equal to :}$$

माना $y = y(x)$ अवकल समीकरण $(1 + y^2)e^{\tan x} dx + \cos^2 x (1 + e^{2\tan x}) dy = 0$ का हल है एवं $y(0) = 1$ है तब $y\left(\frac{\pi}{4}\right)$ बराबर होगा:

- (A) $\frac{2}{e}$ (B) $\frac{2}{e^2}$ (C) $\frac{1}{e^2}$ (D) $\frac{1}{e}$

Ans. D

Sol. $(1 + y^2)e^{\tan x} dx + \cos^2 x (1 + e^{2\tan x}) dy = 0$

$$\frac{dy}{1 + y^2} = -\frac{e^{\tan x} dx}{\cos^2 x (1 + e^{2\tan x})}$$

$$\text{Let } e^{\tan x} = t$$

$$\text{then } e^{\tan x} \cdot \sec^2 x dx = dt$$

$$\Rightarrow \tan^{-1} y = -\frac{dt}{(1 + t^2)}$$

$$\Rightarrow \tan^{-1} y = -\tan^{-1}(t) + C$$

$$\Rightarrow \tan^{-1} y + \tan^{-1}(e^{\tan x}) = C$$

$$y(0) = 1$$

$$\Rightarrow \tan^{-1}(1) + \tan^{-1}(1) = C$$

$$C = \frac{\pi}{2}$$

$$\rightarrow \tan^{-1} y + \tan^{-1}(e^{\tan x}) = \frac{\pi}{2}$$

$$\rightarrow y\left(\frac{\pi}{4}\right) = ?$$

$$\Rightarrow \tan^{-1}(y) + \tan^{-1}(e) = \frac{\pi}{2}$$

$$\Rightarrow \tan^{-1}\left(\frac{y+e}{1-ye}\right) = \frac{\pi}{2}$$

$$\frac{y+e}{1-ye} = \tan\left(\frac{\pi}{2}\right)$$

$$y = \frac{1}{e}$$

112. Let $f(x)$ be a positive function such that the area bounded by $y = f(x)$, $y = 0$ from $x = 0$ to $x = a > 0$ is $e^{-a} + 4a^2 + a - 1$. Then the differential equation, whose general solution is $y = c_1 f(x) + c_2$, where

c_1 and c_2 are arbitrary constants, is :

माना $f(x)$ एक धनात्मक फलन इस प्रकार है कि $y = f(x)$, $y = 0$ द्वारा $x = 0$ से $x = a > 0$ में परिबद्ध क्षेत्रफल $e^{-a} + 4a^2 + a - 1$ है। तब अवकल समीकरण जिसका सामान्य $y = c_1 f(x) + c_2$ है, जहाँ c_1 तथा c_2 स्वेच्छिक अचर है, होगा :

(A) $(8e^x - 1) \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

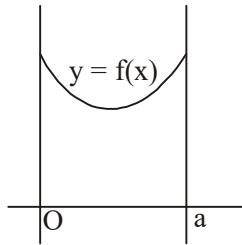
(B) $(8e^x + 1) \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(C) $(8e^x + 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(D) $(8e^x - 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Ans. C

Sol. $\int_0^a f(x) dx = e^{-a} + 4a^2 + a - 1$



$f(a) = -e^{-a} + 8a + 1$

$y = C_1 \left(1 + 8x - \frac{1}{e^x} \right) + C_2$

$\frac{dy}{dx} = C_1 \left(8 + \frac{1}{e^x} \right) \Rightarrow \frac{\left(\frac{dy}{dx} \right)}{\left(8 + \frac{1}{e^x} \right)} = C_1$

$\frac{d^2y}{dx^2} \left(8 + \frac{1}{e^x} \right) - \left(\frac{-1}{e^x} \right) \frac{dy}{dx} = 0$

113. Let $y = y(x)$ be the solution curve of the differential equation $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$, $y(1) = 0$. Then $y(\sqrt{3})$ is equal to :

माना अवकल समीकरा $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$, $y(1) = 0$ का हल वक्र $y = y(x)$ है। तो $y(\sqrt{3})$ बराबर है :

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{12}$

(C) $\frac{\pi}{4}$

(D) $\frac{\pi}{3}$

Ans. C

Sol. $\sec y \frac{dy}{dx} + 2x \sin y = x^3 \cos y$

$$\Rightarrow \sec^2 y \frac{dy}{dx} + 2x \tan y = x^3$$

Let $z = \tan y$

$$\frac{dz}{dx} = \sec^2 y \frac{dy}{dx}$$

$$\Rightarrow \frac{dz}{dx} + 2xz = x^3$$

$$\text{I.F.} = e^{x^2}$$

$$\Rightarrow z \cdot e^{x^2} = \int e^{x^2} \cdot x^3 dx + c$$

$$\Rightarrow \tan y \cdot e^{x^2} = \frac{1}{2} (x^2 e^{x^2} - e^{x^2}) + c$$

$$\Rightarrow \tan(0) \cdot e = \frac{1}{2} (1 \cdot e - e) + c$$

$$\Rightarrow c = 0$$

$$\Rightarrow \tan y = \frac{x^2 - 1}{2}$$

$$f(x) = \tan^{-1} \left(\frac{x^2 - 1}{2} \right) \Rightarrow f(\sqrt{3}) = \frac{\pi}{4}$$

114. The solution curve, of the differential equation $2y \frac{dy}{dx} + 3 = 5 \frac{dy}{dx}$, passing through the point (0, 1) is a conic, whose vertex lies on the line :

अवकल समीकरण $2y \frac{dy}{dx} + 3 = 5 \frac{dy}{dx}$ का हल वक्र, जो (0, 1) से होकर जाता है, एक शांकव है। इस शांकव का शीर्ष निम्न में से किस

रेखा पर है :

- (A) $2x + 3y = 6$ (B) $2x + 3y = 9$ (C) $2x + 3y = -6$ (D) $2x + 3y = -9$

Ans. B

Sol. $2y \frac{dy}{dx} + 3 = 5 \frac{dy}{dx}$

$$2ydy + 3dx = 5dy$$

$$y^2 + 3x = 5y + c, (0, 1)$$

$$1 + 0 = 5 + c$$

$$c = -4$$

$$y^2 - 5y = -3x - 4$$

$$y^2 - 5y + \frac{25}{4} - \frac{25}{4} = 3x - 4$$

$$\left(y - \frac{5}{2}\right)^2 = -3x + \frac{9}{4}$$

$$\left(y - \frac{5}{2}\right)^2 = -3\left(x - \frac{3}{4}\right)$$

$$\left(\frac{3}{4}, \frac{5}{2}\right) \text{ satisfies by } 2x + 3y = 9$$

115. The solution of the differential equation $(x^2 + y^2)dx - 5xy dy = 0, y(1) = 0$, is :

अवकल समीकरण $(x^2 + y^2)dx - 5xy dy = 0, y(1) = 0$ का हल है :

(A) $|x^2 - 4y^2|^5 = x^2$ (B) $|x^2 - 2y^2|^5 = x^2$ (C) $|x^2 - 4y^2|^6 = x$ (D) $|x^2 - 2y^2|^6 = x$

Ans. A

Sol. $(x^2 + y^2)dx = 5xydy$

$$\frac{dy}{dx} = \frac{x^2 + y^2}{5xy}$$

Let $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\frac{1 + v^2}{5v} = v + x \frac{dv}{dx}$$

$$x \frac{dv}{dx} = \frac{1 + v^2}{5v} - v$$

$$x \frac{dv}{dx} = \frac{1 - 4v^2}{5v}$$

$$\frac{5v}{1-4v^2} dv = \frac{dx}{x}$$

$$\int \frac{dx}{x} = -\frac{5}{8} \int \frac{-8v}{1-4v^2} dv$$

$$\ln x = -\frac{5}{8} \ln |1-4v^2| + \ln c$$

$$\ln x + \frac{5}{8} \ln \left| 1 - \frac{4y^2}{x^2} \right| = \ln c \quad \because y(1) = 0$$

$$\text{So } \ln c = 0$$

$$8 \ln x + 5 \ln \left| \frac{x^2 - 4y^2}{x^2} \right| = 0$$

$$8 \ln x + 5 \ln |x^2 - 4y^2| - 10 \ln x = 0$$

$$\ln |x^2 - 4y^2|^5 - \ln x^2 = 0$$

$$|x^2 - 4y^2|^5 = x^2$$

116. Let $\int_0^x \sqrt{1 - (y'(t))^2} dt = \int_0^x y(t) dt$, $0 \leq x \leq 3$, $y \geq 0$, $y(0) = 0$. Then at $x = 2$, $y'' + y + 1$ is equal to :

माना $\int_0^x \sqrt{1 - (y'(t))^2} dt = \int_0^x y(t) dt$, $0 \leq x \leq 3$, $y \geq 0$, $y(0) = 0$ हैं। तो $x = 2$ पर $y'' + y + 1$ बराबर है :

- (A) 1 (B) $\sqrt{2}$ (C) 2 (D) $\frac{1}{2}$

Ans. A

Sol. $\sqrt{1 - (y'(x))^2} = y(x)$

$$1 - \left(\frac{dy}{dx} \right)^2 = y^2$$

$$\left(\frac{dy}{dx} \right)^2 = 1 - y^2$$

$$\frac{dy}{\sqrt{1-y^2}} = dx \quad \text{OR} \quad \frac{dy}{\sqrt{1-y^2}} = -dx$$

$$\Rightarrow \sin^{-1} y = x + c, \sin^{-1} y = -x + c$$

$$x = 0, y = 0 \Rightarrow c = 0$$

$$\sin^{-1}y = x, \text{ as } y \geq 0$$

$$\sin x = y$$

$$\Rightarrow \frac{dy}{dx} = \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\sin x$$

$$\Rightarrow -\sin x + \sin x + 1 = 1$$

117. Let $x = x(t)$ and $y = y(t)$ be solutions of the differential equations $\frac{dx}{dt} + ax = 0$ and $\frac{dy}{dt} + by = 0$ respectively, $a, b \in \mathbb{R}$. Given that $x(0) = 2$; $y(0) = 1$ and $3y(1) = 2x(1)$, the value of t , for which $x(t) = y(t)$, is :

माना अवकल समीकरणों $\frac{dx}{dt} + ax = 0$ तथा $\frac{dy}{dt} + by = 0$, $a, b \in \mathbb{R}$ के हल क्रमशः $x = x(t)$ तथा $y = y(t)$ हैं। यदि $x(0) = 2$; $y(0) = 1$ तथा $3y(1) = 2x(1)$ हैं, तो $x(t) = y(t)$ के लिए t का मान है:

- (A) $\log_3 4$ (B) $\log_{\frac{2}{3}} 2$ (C) $\log_4 3$ (D) $\log_{\frac{4}{3}} 2$

Ans. D

Sol. $\frac{dx}{dt} + ax = 0$

$$\Rightarrow \int \frac{1}{x} dx = -\int a dt$$

$$\Rightarrow \ln|x| = -at + c$$

$$\Rightarrow x = \pm e^c \cdot e^{-at} = \lambda e^{-at}$$

$$\Rightarrow x(0) = 2 \Rightarrow \lambda = 2$$

$$\Rightarrow x = 2e^{-at} \quad \text{_____ (1)}$$

Now, $\frac{dy}{dt} + by = 0$

$$\Rightarrow \int \frac{1}{y} dy = -\int b dt \Rightarrow \ln(y) = -bt + c$$

$$\Rightarrow y = \pm e^c \cdot e^{-bt} = \mu \cdot e^{-bt}$$

$$y(0) = 1$$

$$\therefore \mu = e^{-bt}$$

$$3y(1) = 2x(1)$$

$$3e^{-b} = 2(2e^{-a}) = 4e^{-a}$$

$$e^{-b+a} = \frac{4}{3}$$

$$\Rightarrow x(t) = y(t)$$

$$\Rightarrow e^{-bt} = 2e^{-at}$$

$$\Rightarrow e^{-bt+at} = 2$$

$$\Rightarrow e^{(-b+a)t} = 2$$

$$\Rightarrow \left(\frac{4}{3}\right)^t = 2$$

$$\Rightarrow t \ln \frac{4}{3} = \ln 2$$

$$\Rightarrow t = \log_{\frac{4}{3}} 2$$

118. If $y = y(x)$ is the solution curve of the differential equation $(x^2 - 4)dy - (y^2 - 3y)dx = 0$, $x > 2$, $y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals :

यदि अवकल समीकरण $(x^2 - 4)dy - (y^2 - 3y)dx = 0$, $x > 2$, $y(4) = \frac{3}{2}$ का हल वक्र $y = y(x)$ है तथा इस वक्र की प्रवणता किसी भी x के लिए शून्य नहीं है, तो $y(10)$ का मान बराबर है :

(A) $\frac{3}{1+2\sqrt{2}}$

(B) $\frac{3}{1+(8)^{1/4}}$

(C) $\frac{3}{1-2\sqrt{2}}$

(D) $\frac{3}{1-(8)^{1/4}}$

Ans. B

Sol. $(x^2 - 4)dy - (y^2 - 3y)dx = 0$

$$\Rightarrow \int \frac{dy}{y^2 - 3y} = \int \frac{dx}{x^2 - 4}$$

$$\Rightarrow \frac{1}{3} \int \frac{y - (y-3)}{y(y-3)} dy = \int \frac{dx}{x^2 - 4}$$

$$\Rightarrow \frac{1}{3} (\ln|y-3| - \ln|y|) = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + C$$

$$\Rightarrow \frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + C$$

At $x = 4$, $y = \frac{3}{2}$

$$\therefore C = \frac{1}{4} \ln 3$$

$$\therefore \frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + \frac{1}{4} \ln(3)$$

At $x = 10$

$$\frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{2}{3} \right| + \frac{1}{4} \ln(3)$$

$$\ln \left| \frac{y-3}{y} \right| = \ln 2^{3/4}, \forall x > 2, \frac{dy}{dx} < 0$$

$$\text{as } y(4) = \frac{3}{2} \Rightarrow y \in (0, 3)$$

$$-y + 3 = 8^{1/4} \cdot y$$

$$y = \frac{3}{1 + 8^{1/4}}$$

119. A function $y = f(x)$ satisfies $f(x) \sin 2x + \sin x - (1 + \cos^2 x) f'(x) = 0$ with condition $f(0) = 0$. Then, $f\left(\frac{\pi}{2}\right)$ is equal to :

एक फलन $y = f(x)$ के लिए $f(x) \sin 2x + \sin x - (1 + \cos^2 x) f'(x) = 0$ तथा $f(0) = 0$ हैं। तो $f\left(\frac{\pi}{2}\right)$ बराबर है :

- (A) 2 (B) 1 (C) 0 (D) -1

Ans. B

Sol.
$$\frac{dy}{dx} - \left(\frac{\sin 2x}{1 + \cos^2 x} \right) y = \frac{\sin x}{1 + \cos^2 x}$$

$$\text{If } = e^{-\int \frac{\sin 2x}{1 + \cos^2 x} dx} = e^{\ln(1 + \cos^2 x)} = 1 + \cos^2 x$$

So general solution of DE

$$y(1 + \cos^2 x) = \int \frac{\sin x}{1 + \cos^2 x} \cdot (1 + \cos^2 x) dx$$

$$y(1 + \cos^2 x) = -\cos x + c$$

$$y(1 + \cos^2 x) + \cos x = c$$

were $y(0) = 0$

$$0 + 1 = c$$

$$y(1 + \cos^2 x) + \cos x = 1$$

$$y = \frac{1 - \cos x}{1 + \cos^2 x}$$

$$y\left(\frac{\pi}{2}\right) = \frac{1 - 0}{1 + 0} = 1$$

120. If $\sin\left(\frac{y}{x}\right) = \log_e |x| + \frac{\alpha}{2}$ is the solution of the differential equation $x \cos\left(\frac{y}{x}\right) \frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then α^2 is equal to :

यदि अवकल समीकरण $x \cos\left(\frac{y}{x}\right) \frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x$, $y(1) = \frac{\pi}{3}$, का हल $\sin\left(\frac{y}{x}\right) = \log_e |x| + \frac{\alpha}{2}$ है, तो

α^2 बराबर है :

(A) 3

(B) 9

(C) 12

(D) 4

Ans. A

Sol. Given differential equation is $x \cos \frac{y}{x} \frac{dy}{dx} - y \cos \frac{y}{x} = x$

$$\Rightarrow \cos \frac{y}{x} \left(x \frac{dy}{dx} - y \right) = x$$

$$\Rightarrow \cos \frac{y}{x} \left(\frac{x \frac{dy}{dx} - y \cdot 1}{x^2} \right) = \frac{1}{x}$$

Let $\frac{y}{x} = t$

$$\cos t \frac{dt}{dx} = \frac{1}{x}$$

$$\cos t dt = \frac{1}{x} dx$$

on integration $\sin t = \ln |x| + c$

$$y(1) = \frac{\pi}{3} \Rightarrow \sin \frac{\pi}{3} = 0 + c$$

$$\Rightarrow c = \frac{\sqrt{3}}{2}$$

$$\therefore \text{ solution is } \sin \frac{y}{x} = \ln|x| + \frac{\sqrt{3}}{2}$$